

Ismail AISSAOUI

State Engineer in Artificial Intelligence

ismail.aissaoui.pro@gmail.com

+213 660 70 77 96

ismail-aissaoui.vercel.app

To the PhD Selection Committee
EDT Program - INRAE / Université Côte d'Azur
Attn: Dr. Lorenzo Sala and Prof. Julien Deantoni

May 1, 2026

Subject: Application for PhD: Compositional Hybrid Digital Twins for Fermented Microbial Ecosystems (Rank 7 - PC1)

Dear Dr. Sala and Prof. Deantoni,

As a State Engineer in Artificial Intelligence from ESI-SBA, I am writing to express my enthusiastic application for the PhD position in "Compositional Hybrid Digital Twins for Fermented Microbial Ecosystems" within the Engineering Digital Twin (EDT) national program. This application is motivated by the alignment between my work on hybrid modeling and the interdisciplinary nature of your project.

Currently, I am completing my thesis at Sonatrach, where I have developed a **Physics-Informed Digital Twin** for gas turbine monitoring. While my current application is industrial, the core of my research is the **hybridization of mechanistic models (thermodynamics) with data-driven surrogates (Mamba-SSM/xLSTM)**. I have spent significant effort designing custom loss functions that enforce physical consistency, an experience that has given me a deep understanding of the mathematical challenges involved in blending first-principles equations with partially understood observed dynamics.

I am particularly drawn to your focus on **compositional hybrid modeling operators** and reusable hybridization patterns. I am eager to transition my expertise in "Hybrid Model Forge" from industrial energy systems to the complex, non-linear world of microbial fermentation. My technical background in **ODE-based modeling** and high-dimensional statistical analysis aligns perfectly with your goal of developing multi-fidelity digital twins that can handle incomplete observations and evolving biological knowledge.

My background in hybrid modeling and my experience with production-grade PyTorch and scientific computing (Python/Julia) provide me with the necessary tools to contribute to the **Artemis platform** and the broader EDT ecosystem. I am a highly autonomous researcher, comfortable bridging the gap between computational biology and industrial-scale digital twin engineering.

Thank you for your time and consideration. I look forward to the possibility of discussing how my experience in physics-aware AI and dynamical systems can contribute to the success of your research group at INRAE and UniCA.

Sincerely,

Ismail AISSAOUI